

Quantifying Doneness.

An internal dashboard for the team behind GRILL X. It makes the **AI's doneness read visible**, and confirms it stays consistent on every cut.

AI Doneness
Sensing

Validation
Dashboard

Beyond Honeycomb
R&D



Cooking is **science**, so we measure it.

Founded in Seoul in 2020 by a team of ex-Samsung Research engineers, **Beyond Honeycomb** builds AI cooking robots that quantify flavor and reproduce chef-level results on every cut.

100+

customer brands
(Nov 2025)

150+

GRILL X robots
deployed

\$14.1M

raised, led by
LB Investment

-80%

staff training time
(operator-
reported)



The world's first **AI grilling** robot.

GRILL X grills to chef-level doneness on its own. It reads each cut as it cooks and adjusts in real time, so the result is the same no matter who works the line. One button, and it fits onto a kitchen's existing grill.

up to 80 servings / hour

consistent every shift



Senses as it cooks

Judges doneness from the meat itself, not from a fixed timer or temperature.



Learns from chefs

Captures an expert's technique and reproduces it on every shift.



Built in-house

Sensing, on-device AI and robotics are developed together, and clamp onto any grill.



Proven at scale

NAVER and KT cafeterias, plus Andaz Seoul, Korea's first 5-star-hotel AI chef robot.

Can you trust what the AI **sees**?

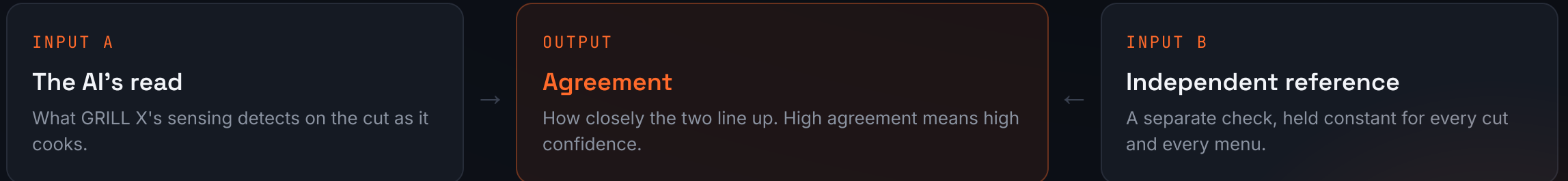
GRILL X decides doneness from its own read of the meat. For the team building it, that raises a practical question. When the AI makes the call, how do we see what it saw, and confirm the read holds up cut after cut?



Doneness is decided in seconds, inside the machine. The dashboard brings that decision into view, so the team can check it instead of trusting it blind.

Cross-check the read. **Measure the agreement.**

There is no perfect answer key for a steak, so the tool never crowns a winner. It compares the AI's read against an independent reference and measures how closely the two line up. The closer they agree, the more we can trust the call.



Agreement, not accuracy. We never claim one read is correct. We show how much the AI agrees with an independent check, and where it does not. That is an honest signal the team can act on.

Agreement you can see at a glance.

For any cut, the map paints where the AI's read and the reference line up, and where they pull apart. Strong overlap means high confidence. The gaps are exactly what the team reviews.

ON THIS CUT

Strong agreement

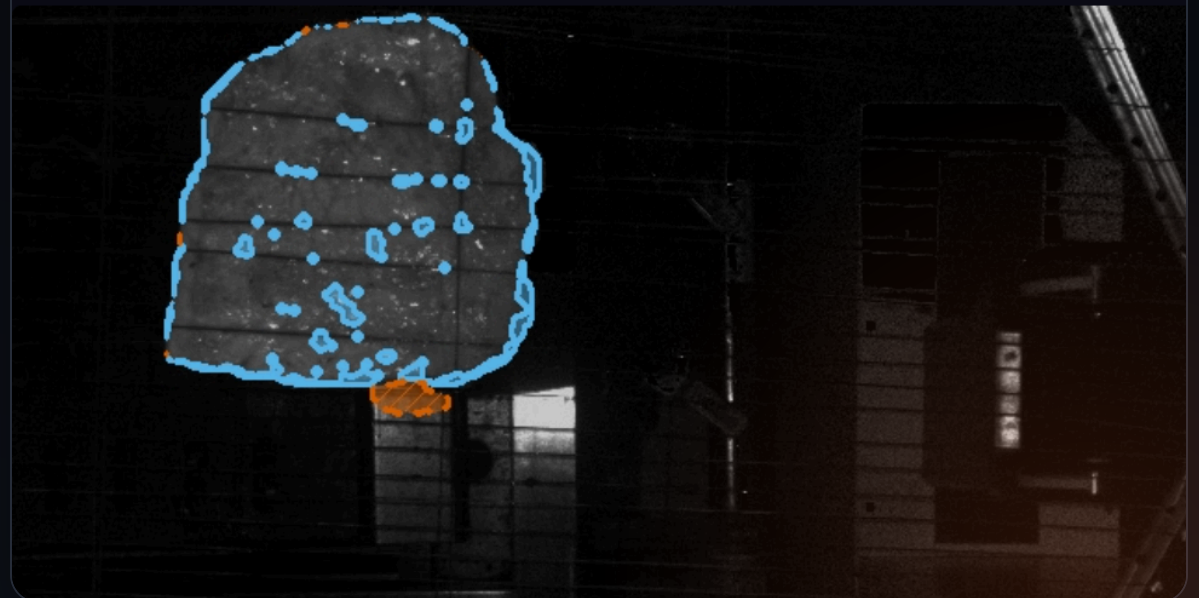
The read holds against the reference, with only the edges left to inspect.

Both agree AI read Reference

Demo mode — curated capture: device 206 beef/striploin 250903 pos 1.

1. Hook — a pitmaster judges char on the charbroiler by eye. Quantify it across 11 wavelengths.
2. Frame — the mono led10 (740nm) grayscale frame is what the sensor sees.
3. Hero reveal — where AI segmentation (cyan) and the rule ROI (vermillion) diverge.
4. The one number — IoU and signed area delta. Same rules, cleaner AI ROI, no tuning.
5. Why it matters — 11 wavelengths: AI segments, rules decide.
6. Close — no ground-truth labels, so this shows agreement, not accuracy.

ROI-diff overlap-map



One workspace, from headline to single cut.

Demo mode — curated capture: device 206 beefstation 250003 pos 1.

1. Hook — a pitmaster judges char on the charcoal by eye. Quantify it across 11 wavelengths.
2. Frame — the mono led10 (740nm) grayscale frame is what the sensor sees.
3. Hero reveal — where AI segmentation (cyan) and the rule ROI (vermillion) diverge.
4. The one number — IOI and signed area delta. Same rules, clearer AI ROI, no tuning.
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ROI-diff overlap-map



Compare

The agreement map up top, with a confidence read for the cut.

signal (mono led10) AI ROI overlay Rule ROI overlay



Rule ROI



Doneness

- B BURNT
- S SLIGHTLY BURNT
- P PROPER
- NO NOT_DONE

Per-panel

Side-by-side views of one cut, broken down by doneness.

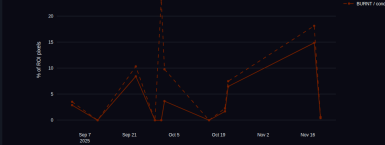
BH Sensing

At 100 vs Rule 100 — agreement, not accuracy

Mode

Menu

By-date trends



Trends

How doneness moves over time, read by read.

BH Sensing

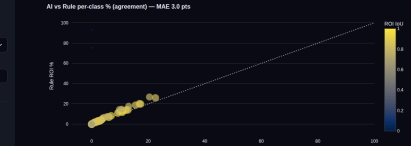
At 100 vs Rule 100 — agreement, not accuracy

Mode

Menu

AI vs Rule (agreement)

At vs Rule per class % (agreement) — MAE 5.0 pts



Explore

Filter and sort every capture, then open any one in detail.

A guided **Demo mode** walks through a single cut step by step. **Explore mode** opens the full set of captures to filter and compare.

A read the team can **trust**.

At a glance

Confidence on every cut, without scrolling through raw captures.

Catches drift

A shift in the read shows up early, before it reaches a plate.

Stays consistent

The read holds steady as menus and cuts change.

Built to grow

Ready for new proteins as GRILL X expands.

The takeaway

The AI's read holds up against an independent check, and the places it diverges are now visible instead of a matter of opinion. The team can trust the call, and sharpen it where it counts.

— BUILT WITH

Small stack, runs anywhere.

Everything runs on an ordinary laptop, with no cloud and no GPU.

ONNX Runtime

Runs the AI model on a regular CPU.

DuckDB

Fast local store for date and menu rollups.

Streamlit

The dashboard itself, read-only over the results.

Plotly

Charts on a colorblind-safe palette.

OpenCV · NumPy

Image handling and the number work underneath.

Python · uv

Pinned dependencies, reproducible setup.

Built to stay honest

The tool never calls one read "accurate" or "better." It reports agreement. Differences are shown plainly, with no color taking sides, and the whole palette is colorblind-safe.

— WHERE THIS GOES

An honest read on what the **AI sees.**

- **Trust the call.** Confidence in the AI's doneness read, at a glance.
- **Catch drift early.** See a change in the read before it reaches a plate.
- **Grow with the menu.** Ready as GRILL X moves to new proteins.



BEYOND HONEYCOMB

Chef-level grilling with AI precision, every time.

